Cheese Moon – Teacher notes

This task has students practise the previous lessons (shapes, fill and stroke, Node tool, path operations) on their own. Print it or share it – students work independently while you circulate. Shortcuts are left out on purpose: every action goes through a menu or a toolbar, so students learn where things live; students who already know Ctrl+Z or Ctrl+D may of course use them. Exact values (stroke widths, ratios, alignment) are left out too – everything is done by eye, so the time goes into the operations.

Timing

Minutes	Phase	Watch for
0–2	Open, save with name	Has everyone saved before drawing?
2–6	Sky + stars	Star settings are in the tool controls bar; stop them drawing 30 stars
6–13	Cheese Moon	The longest step. Union FIRST, then Difference with exactly 2 objects
13–18	Rocket body	Intersection: only the shared part stays
18–24	Fins	Object to Path before the Node tool; Duplicate + Flip Horizontal
24–28	Porthole	Ring = Difference
28–32	Flame, order	Lower vs. Lower to Bottom
32–35	Save, export, upload	SVG + PNG

Common mistakes and the quick fix

- Difference removes the Moon instead of the craters: the Moon was on top. Edit ▸ Undo, select the craters, Object ▸ Raise to Top, repeat.
- Difference is greyed out: not exactly two objects are selected. Read the status bar.
- The shape “vanished” after a Path operation: the result takes the bottom object's style – it may be sky-coloured with no stroke. Set the colour.
- The Node tool shows round or square handles, not corner nodes: the rectangle is not a path yet. Path ▸ Object to Path.
- The flame vanished after Lower to Bottom: it is behind the sky. Edit ▸ Undo, or open Layer ▸ Layers and Objects… (older versions: Object ▸ Objects…), click the triangle in the list to select it, then Object ▸ Raise.
- The porthole hole is not cut: the small circle was drawn first and sits underneath. Select it, Object ▸ Raise to Top, then Difference.
- Path operations do not work on a group: Object ▸ Ungroup first.

Assessment (10 points)

Element	Points
Sky + stars (fill and stroke set correctly)	1
Moon with craters and a bitten edge (Union + Difference)	3
Rocket body with a flat bottom (Intersection)	2
Symmetrical fins (Node tool + flip)	2
Ring porthole (Difference)	1
Flame in the right order, SVG + PNG handed in	1

In the SVG you can check that the Moon really was made with path operations: the cratered Moon is a single `path` element, not a pile of circles. Differentiation: weaker students can stop at 3 craters; fast students get the bonus table.